Rithvik Jami

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.76
Sri Chaitanya	Class XII (TSBIE)	2023	98.30%
Sri Chaitanya	Class X (SSC)	2021	10.00%

EXPERIENCE

Warner Bros. Discovery : SWE Intern

Engineered a local fallback for Warner Bros.' GCP-based closed-captioning pipeline, **preserving core functionality** during production/cloud failures.

- Developed the end-to-end workflow covering from **raw audio processing to final subtitle generation along with multiple stages in between**, while laying the foundation for future multi-cloud migration by using **Metaflow**.
- **Evaluated and compared ML models** for local execution, analyzing the trade-off between **accuracy and execution time** to ensure practical performance on local infrastructure.

Research on *Parallel community detection in large graphs* : Prof. Sathya Peri

- **Optimized parallel Louvain community detection for large-scale graphs** by reusing graph coloring across iterations and implementing **persistent thread-pool execution**, reducing synchronization overhead and eliminating mutex contention to achieve up to **4× speedup** over the baseline implementation.

SKILLS

Programming Languages : C/C++, Python, Javascript, Assembly(RISC-V)

Web-Technologies : HTML, CSS/Tailwind-CSS, Express.js*, Node.js*

Technologies : MySQL, MongoDB, Postgres, Flex, Bison

Development Tools and Platforms : Git, GitHub, Make, Docker, LaTeX, Bash, Linux

CP : Solved 200+ problems on Leetcode, Rated 1114 on Codeforces

POSITIONS OF RESPONSIBILITY

Blockspace Core - (Blockchain and Web3 club of IITH)

- Explored blockchain fundamentals and smart contract development using Solidity, and contributed to technical sessions and knowledge-sharing initiatives for the wider IITH community.

Web-Dev Core,Media Council IIT Hyderabad

- Contributed to the maintenance and developnment of the official IIT Hyderabad website.

RELEVANT COURSES

Computer Science Courses :

Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Distributed Computing, Computer Architecture, Theory of Computation, Compilers, Software Engineering, Parallel & Concurrent Programming

Mathematics & AI :

Discrete Mathematics, Probability and Random Variables, Linear Algebra, Convex Optimizations, Foundations of ML, Deep Learning

PROJECTS

Centralized Communication Library

- **Architected a modular full-stack platform** to centralize content creation, review, and publishing across teams.
- Implemented **content lifecycle management** with versioning, rollback, RBAC, and quorum-based approval workflows.
- **Integrated semantic search** using Elasticsearch and vector embeddings, with **Prisma and PostgreSQL** for structured data, **MongoDB** for document-based data, and **Docker Compose + Nginx** for service orchestration and deployment.

AFT (Another frequency thing) Compiler

- Engineered a **statically typed DSL compiler in C++**, covering lexical analysis, parsing, AST construction, and LLVM IR generation.
- Implemented **semantic analysis and LLVM-based code generation** with support for functions, control flow, and type checking.
- Added **symbol table management and LLVM optimization support** for efficient code generation.

Kademlia Distributed Hash Table

- Built a **peer-to-peer DHT in C++ using UDP**, supporting decentralized discovery, storage, and retrieval.
- Implemented **Kademlia routing, iterative lookups, key-value storage, and peer communication**.
- Designed the system for **fault-tolerant peer communication and scalable distributed network operations**.

Multi-Client Quiz system

- Built a multithreaded client-server quiz system in **C++** using raw **TCP** sockets, with cached and non-cached server and **LLM** generated questions .
- Designed a **custom HTTP-inspired application-layer protocol** for structured client-server communication over TCP.
- Analyzed network performance using **packet-capture tools**, measuring **throughput** and **latency** across multi-client scenarios.

EXTRACURRICULAR ACTIVITIES AND VOLUNTEERING WORK

National Service Scheme (NSS) - Student Volunteer